

Nithish Krishnabharathi GNANI

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WORK EXPERIENCE

Indian Institute of Science

Technical Associate

Bangalore, India

2019-2024

- Led industry-funded research projects. I was sponsored by the Center for Networked Intelligence (CNI), a Cisco CSR initiative at Robert Bosch Center for Cyber-Physical Systems (RBCCPS). (Promoted from Research Assistant to this role.)

Coca-Cola (HCCBPL)

Team Leader, Maintenance

Hyderabad, India

2017-2018

- Managed a team of 20+ associates to achieve consistent reliability of utility systems.

EDUCATION

Polytech Nice Sophia, Université Cote d'Azur

KTH Royal Institute of Technology

EIT Digital Dual Master of Science - Autonomous Systems and Intelligent Robots

Minor: Innovation and Entrepreneurship

Sophia Antipolis, France

Stockholm, Sweden

2024 - 2026

National Institute of Technology Karnataka (NITK)

Bachelor of Technology - Mechanical Engineering

Surathkal, Karnataka, India

2013 - 2017

SELECTED PROJECTS (URL for more: <https://nithishkgnani.github.io/projects/>)

Designing Tactile Cyber-Physical Systems

Mar 2022 - Aug 2024

- Created a testbed for real-time human-robot interaction with ultra-reliable low latency communication. Built a remote robotic arm system interfaced with a haptic device for teleoperation across continents.
- Implemented IEEE 802.1 TSN features on hardware for enhanced network reliability and latency.

5G AMMAZING

Mar 2020 - Jun 2022

- Developed a 5G mmWave system for infotainment with split data and control planes over 5G and Wi-Fi.
- Won first place (€6,500 cash award) in 5G Hackathon 2020 by Government of India among 1100+ teams.

Acoustic PUF for Sensor Security

Nov 2019 - Jul 2021

- Identified and localized sensor devices using acoustic fingerprinting for security and integrity monitoring.

Energy Harvesting IoT Sensors

Nov 2019 - Jun 2020

- Developed adaptive sampling algorithms for energy-efficient data management in IoT sensors.

Total Productive Maintenance

Nov 2017 - Apr 2018

- Improved reliability and reduced failures in Coca-Cola's bottling plant through TPM steps.

SKILLS & INTERESTS

Programming : Python, MATLAB, Simulink, P4, C, C++, Embedded C, Shell
Hardware : Netronome Agilio Smart NIC, Nordic nRF52 series, Arduino
Tools : Linux, Git, ROS, Docker, LaTeX, Autodesk Fusion 360
Creative : 3D printing, Shotcut, Snapseed
Languages : English (IELTS band score 8/9), French (A1), Hindi, Tamil, Kannada, Sourashtra



PUBLICATIONS

EdgeP4: In-Network Edge Intelligence for a Tactile Cyber-Physical System Testbed Across Cities

Best Paper Award

Nithish K Gnani, Joydeep P, Deepak C, Himanshu V, Soumya R, Kaushal M, T. V. Prabhakar, C Singh

IEEE International Conference on Computer Communications Workshops (INFOCOM) 2024

IEEE INFOCOM '24

Towards a TSN-DetNet Intercity Testbed for Tactile Cyber-Physical Systems

Joydeep P, Nithish K Gnani, Deepak C, Chandramani Singh, T. V. Prabhakar, H. K. Atluri, Paventhan A

IEEE International Conference on Computer Communications Workshops (INFOCOM) 2024

IEEE INFOCOM '24

Sensor Identification via Acoustic Physically Unclonable Function

Girish Vaidya, T. V. Prabhakar, Nithish K Gnani, Ryan Shah, Shishir Nagaraja

Digital Threats: Research and Practice, Volume 4, Issue 2

ACM DTRAP journal

Judicious data management for sustaining an energy harvesting sensor node

K Singh, P Shukla, Sachin S. M., Nithish K Gnani, Prabhakar T. V., Joy Kuri

Concurrency and Computation: Practice and Experience, Volume 33, Issue 23

Wiley CCPE journal